

6. Continuous distributions

Recall: $(\Omega, \mathcal{F}, \mathbb{P})$ is our probability space

$X: \Omega \mapsto \mathbb{R}$ is a random variable

The distribution function of X is $F(x) = \mathbb{P}(X \leq x)$.

properties: a) F is monotonically increasing (often not strictly)

b) $\lim_{x \rightarrow -\infty} F(x) = 0$

c) $\lim_{x \rightarrow \infty} F(x) = 1$

d) F is left continuous, hence $\forall x_0 \in \mathbb{R}, \lim_{x \rightarrow x_0^-} F(x) = F(x_0)$

We only used the monotonicity and continuity of $\mathbb{P}$ in the proof.

Note: If F is a function satisfying the ~~(a)-(d)~~ properties above (a)-(d), then F is a distribution function of some random variable.

I. Continuous random variables

def: We call a random variable X continuous if the distribution function of X ($x \mapsto F(x)$) is continuous.

def: We call the random variable X absolutely continuous if $\exists f: \mathbb{R} \mapsto \mathbb{R}$ such that $F(x) = \int_{-\infty}^x f(x) dx$

We call f the density function of X .

The following proposition follows from properties a) and c) of F .

Prop: If X is absolutely continuous with density function f , then

a) $f(x) \geq 0, \forall x \in \mathbb{R}$ and b) $\int_{-\infty}^{\infty} f(x) dx = 1$

Note: If f satisfies these properties, then f is a density function.

Prop: If X is absolutely continuous, then $\forall x_0 \in \mathbb{R}, \mathbb{P}(X = x_0) = 0$.

proof: Let f be the density function of f , then

$$\mathbb{P}(X = x_0) = \int_{x_0}^{x_0} f(x) dx = 0 \quad \square$$

It implies $\forall a \in \mathbb{R}, \mathbb{P}(X < a) = \mathbb{P}(X \leq a)$, thus

$$\begin{aligned} \mathbb{P}(a \leq X < b) &= \mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a < X < b) = \\ &= \mathbb{P}(X < b) - \mathbb{P}(X < a) = \int_{-\infty}^b f(x) dx - \int_{-\infty}^a f(x) dx = \int_a^b f(x) dx \quad \text{for any } a, b \in \mathbb{R} \end{aligned}$$

Example: Let X be absolutely continuous with density function

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$c = ?$$

$$P(X > 1) = ?$$

$$1 = \int_{-\infty}^{\infty} f(x) dx = \int_0^2 c(4x - 2x^2) dx = \left[c(2x^2 - \frac{2}{3}x^3) \right]_0^2 = c(8 - \frac{16}{3}) = c \cdot \frac{8}{3} \Rightarrow \underline{\underline{c = \frac{3}{8}}}$$

$$\begin{aligned} \underline{\underline{P(X > 1)}} &= 1 - P(X \leq 1) = 1 - \int_{-\infty}^1 f(x) dx = 1 - \int_0^1 \frac{3}{8}(4x - 2x^2) dx = \\ &= 1 - \left[\frac{3}{8}(2x^2 - \frac{2}{3}x^3) \right]_0^1 = 1 - \frac{3}{8}(2 - \frac{2}{3}) = 1 - \frac{3}{8} \cdot \frac{4}{3} = \underline{\underline{\frac{1}{2}}} \end{aligned}$$

It follows from the definition, that $P(X \in [a, a + \epsilon)) = \int_a^{a+\epsilon} f(x) dx = f(a) \cdot \epsilon + o(\epsilon)$.
That is $f(a)$ tells us how dense X takes values around $a \in \mathbb{R}$.

def. Let X be an absolutely continuous random variable with density function f . Then the expected value of X is

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx \quad \text{if} \quad \int_{-\infty}^{\infty} |x| f(x) dx < \infty.$$

Otherwise it does not exist, however:

$$\text{If } \int_{-\infty}^{\infty} |x| f(x) dx = \infty \text{ but } \int_{-\infty}^0 |x| f(x) dx < \infty, \text{ then we say } E[X] = \infty.$$

$$\text{If } \int_{-\infty}^{\infty} |x| f(x) dx = \infty \text{ but } \int_0^{\infty} |x| f(x) dx < \infty, \text{ then we say } E[X] = -\infty$$

Example. Let X be a random variable with density $f(x) = \frac{1}{\pi(1+x^2)}$.

$$\int_{-\infty}^{\infty} \frac{1}{\pi(1+x^2)} dx = \left[\frac{1}{\pi} \arctan(x) \right]_{-\infty}^{\infty} = \frac{1}{\pi} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = 1$$

What is the expected value of X ? ↳ It is indeed a density function.

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx = \int_{-\infty}^0 \frac{x}{\pi(1+x^2)} dx + \int_0^{\infty} \frac{x}{\pi(1+x^2)} dx = \\ &= \int_0^{\infty} \frac{x}{\pi(1+x^2)} dx - \int_0^{\infty} \frac{x}{\pi(1+x^2)} dx = \left[\frac{1}{2\pi} \ln(1+x^2) \right]_0^{\infty} - \left[\frac{1}{2\pi} \ln(1+x^2) \right]_0^{\infty} = \infty - \infty \end{aligned}$$

$\Rightarrow$ No value can be assigned as the expectation of X !

Prop.: If X is a non-negative absolutely continuous random variable, then

$$E[X] = \int_0^{\infty} P(X > y) dy$$

proof:

$$\int_0^{\infty} P(X > y) dy = \int_0^{\infty} \int_y^{\infty} f(x) dx dy = \int_0^{\infty} \int_0^x f(x) dy dx = \int_0^{\infty} x f(x) dx = E[X] \quad \square$$

X has a density f

Fubini's Theorem

Prop. Let X be an absolutely continuous random variable with density function f . If $g: \mathbb{R} \rightarrow \mathbb{R}$ and $Y = g(X)$, then

$$E[Y] = E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Prop. Let X be an absolutely continuous random variable, then

$$\forall a, b \in \mathbb{R} : E[aX + b] = aE[X] + b.$$

If Y is also absolutely continuous, then

$$E[X] + E[Y] = E[X + Y].$$

def. The variance of an absolutely continuous random variable X is

$$D^2[X] = \text{Var}(X) = E[(X - E[X])^2]$$

The standard deviation of X is $D[X] = \sqrt{D^2[X]}$

Prop. For an absolutely continuous random variable X

$$D^2[X] = E[X^2] - E[X]^2.$$

def. Let X be an absolutely continuous random variable.

The median of X is $F^{-1}(1/2)$. $\hookrightarrow$ it is the point $x_0 \in \mathbb{R}$, $P(X < x_0) = P(X > x_0)$

The r-quantile of X is $Q(r) = F^{-1}(r) \hookrightarrow x_0 \in \mathbb{R}$, $P(X \leq x_0) = r$ & $P(X > x_0) = 1 - r$

The mode of X is where the maximum of the density function is attained.

II. Uniform distribution

def. Let $\alpha < \beta$ be real numbers. We say that X is uniformly distributed on $[\alpha, \beta]$ if its density is

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & , x \in [\alpha, \beta] \\ 0 & , \text{otherwise} \end{cases} \Rightarrow F(x) = \begin{cases} 1 & , x > \beta \\ \frac{x - \alpha}{\beta - \alpha} & , \alpha < x \leq \beta \\ 0 & , x \leq \alpha \end{cases}$$

Notation, $X \sim \text{Uni}[\alpha, \beta]$

Prop. If $X \sim \text{Uni}[\alpha, \beta]$, then $E[X] = \frac{\beta + \alpha}{2}$ and $D^2[X] = \frac{(\beta - \alpha)^2}{12}$

proof.

$$E[X] = \int_{\alpha}^{\beta} \frac{x}{\beta - \alpha} dx = \left[\frac{x^2}{2(\beta - \alpha)} \right]_{\alpha}^{\beta} = \frac{\beta^2 - \alpha^2}{2(\beta - \alpha)} = \frac{\beta + \alpha}{2}$$

$$E[X^2] = \int_{\alpha}^{\beta} \frac{x^2}{\beta - \alpha} dx = \left[\frac{x^3}{3(\beta - \alpha)} \right]_{\alpha}^{\beta} = \frac{\beta^3 - \alpha^3}{3(\beta - \alpha)} = \frac{\beta^2 + \alpha\beta + \alpha^2}{3}$$

$$\begin{aligned} \Rightarrow D^2[X] &= E[X^2] - E[X]^2 = \frac{\beta^2 + \alpha\beta + \alpha^2}{3} - \frac{(\beta + \alpha)^2}{4} = \frac{4\beta^2 + 4\alpha\beta + 4\alpha^2 - 3\beta^2 - 6\alpha\beta - 3\alpha^2}{12} = \\ &= \frac{\beta^2 - 2\alpha\beta + \alpha^2}{12} = \frac{(\beta - \alpha)^2}{12} \quad \square \end{aligned}$$

Prop. Let $F: \mathbb{R} \mapsto [0, 1]$ be a continuous and strictly monotonically increasing function that satisfies properties a-d of distribution functions.

Let $X \sim \text{Uni}[0, 1]$ and $Y = F^{-1}(X)$. Then the distribution function of Y is F .

proof. $P(Y < y) = P(F^{-1}(X) < y) = P(X < F(y)) = \frac{F(y) - 0}{1 - 0} = F(y) \quad \square$

Prop. If $X \sim \text{Uni}[\alpha, \beta]$, then $X = (\beta - \alpha)Y + \alpha$, where $Y \sim \text{Uni}[0, 1]$.

proof.

$$P(X < x) = P\left(\underbrace{\frac{X - \alpha}{\beta - \alpha}}_{= Y} < \underbrace{\frac{x - \alpha}{\beta - \alpha}}_{= y}\right) = \begin{cases} 1, & x > \beta \Leftrightarrow y > 1 \\ \frac{x - \alpha}{\beta - \alpha}, & \alpha < x \leq \beta \Leftrightarrow 0 < y \leq 1 \\ 0, & x \leq \alpha \Leftrightarrow y < 0 \end{cases} \quad \square$$

III. Exponential distribution

def. We say that a random variable X has exponential distribution with parameter λ if its distribution and density functions are respectively $(\lambda > 0)$

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0. \end{cases}$$

$f(x)$ is indeed a density, $\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_0^{\infty} = \underline{\underline{1}}$.

Notation. $X \sim \text{Exp}(\lambda)$

Prop. If $X \sim \text{Exp}(\lambda)$, then $E[X] = D[X] = 1/\lambda$.

proof.

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \lambda e^{-\lambda x} dx \stackrel{\text{integration by parts, } u(x)=x, v'(x)=\lambda e^{-\lambda x}}{=} [x \cdot (-e^{-\lambda x})]_0^{\infty} - \int_0^{\infty} 1 \cdot (-e^{-\lambda x}) dx = \frac{1}{\lambda} \underbrace{\int_0^{\infty} \lambda e^{-\lambda x} dx}_{=1} = \underline{\underline{\frac{1}{\lambda}}}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \stackrel{\text{integration by parts, } u(x)=x^2, v'(x)=\lambda e^{-\lambda x}}{=} [x^2 \cdot (-e^{-\lambda x})]_0^{\infty} - \int_0^{\infty} 2x \cdot (-e^{-\lambda x}) dx = \frac{2}{\lambda} \underbrace{\int_0^{\infty} x \lambda e^{-\lambda x} dx}_{=E[X]} = \underline{\underline{\frac{2}{\lambda^2}}}$$

$$\Rightarrow D^2[X] = E[X^2] - E^2[X] = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \Rightarrow \underline{\underline{D[X] = \frac{1}{\lambda}}} \quad \square$$

Prop. The exponential distribution is renewable ("forever young"), that is if

$X \sim \text{Exp}(\lambda)$, then

$$\forall t, s \in \mathbb{R}_+ : P(X \geq t+s | X \geq s) = P(X \geq t)$$

proof:

$$P(X \geq t+s | X \geq s) = \frac{P(X \geq t+s) \cap \{X \geq s\}}{P(X \geq s)} = \frac{P(X \geq t+s)}{P(X \geq s)} =$$

$$= \frac{1 - F(t+s)}{1 - F(s)} = \frac{1 - (1 - e^{-\lambda(t+s)})}{1 - (1 - e^{-\lambda s})} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = 1 - F(t) = P(X \geq t) \quad \square$$

Thm: The exponential distribution is the only continuous distribution with the renewal property.

proof: Let X be a continuous random variable with distribution function F . Suppose that X has the renewal property. It means

$$P(X \geq t+s | X \geq s) = \frac{P(X \geq t+s)}{P(X \geq s)} = P(X \geq t)$$

$$g(t+s) = P(X \geq t+s) = P(X \geq t) P(X \geq s) = g(t) g(s), \text{ where}$$

$g(t) := P(X \geq t) = 1 - F(t)$. Let $m, n \in \mathbb{N}$, $n \neq 0$, then

$$\underset{\text{renewal}}{g\left(\frac{m}{n}\right)} = \underset{n\text{-many}}{g\left(\underbrace{\frac{m}{n} + \dots + \frac{m}{n}}_m\right)} = g(m) = \underset{m\text{-many}}{g\left(\underbrace{1 + \dots + 1}_m\right)} = \underset{\text{renewal}}{g(1)^m}$$

$$\Rightarrow g\left(\frac{m}{n}\right) = g(1)^{\frac{m}{n}}$$

By definition, g is monotonically decreasing.

Choose an arbitrary $t \in \mathbb{R}_+$, then

$\exists \{q_n\}_{n \geq 1}, \{p_n\}_{n \geq 1} \subseteq \mathbb{Q}$ sequences such that

$$q_n \xrightarrow[n \rightarrow \infty]{} t, p_n \xrightarrow[n \rightarrow \infty]{} t \text{ and } \forall n \in \mathbb{N}, q_n < t < p_n.$$

$$\Rightarrow \underset{g(1)}{g(q_n)} = g(p_n) \leq g(t) \leq g(q_n) = \underset{g(1)}{g(p_n)} \quad \begin{array}{l} \text{squeeze theorem} \\ \Rightarrow \forall t \in \mathbb{R}_+, g(t) = g(1)^t \end{array}$$

$\uparrow$ we showed it for rational numbers $\nwarrow$ g is mon. decreasing

Since $0 < g(1) < 1$ we may write $g(t) = e^{-\lambda t}$, $\forall t \in \mathbb{R}_+$ with $\lambda = -\ln g(1)$.

$$\Rightarrow e^{-\lambda t} = g(t) = 1 - F(t) \Rightarrow F(t) = 1 - e^{-\lambda t}, \forall t \in \mathbb{R}_+ \quad \square$$

Example: On a train station the waiting time between 2 trains is 5 minutes on average. I have been waiting for 7 minutes already, and didn't see any trains. What is the probability that altogether I need to wait 10 mins if we model the waiting time with exponential distribution?

$$X \sim \text{time I need to wait (min)} \quad E[X] = 5 \Rightarrow \lambda = \frac{1}{5}$$

$$X \sim \text{Exp}(\lambda)$$

$$P(X \geq 10 | X \geq 7) = P(X \geq 3) = 1 - (1 - e^{-\frac{1}{5} \cdot 3}) = e^{-\frac{3}{5}}$$

IV. Transforming random variables

Example: Let $X \sim \text{Uni}[-1, 1]$. What is the distribution of X^2 ?

Write $Y = X^2$ and let F_Y, f_Y be its distribution and density ^{functions respectively} ~~function respectively~~.

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = \begin{cases} 0, & y < 0 \\ \textcircled{*}, & 0 \leq y < 1 \\ 1, & 1 \leq y \end{cases} \quad \text{for } y \in \mathbb{R}$$

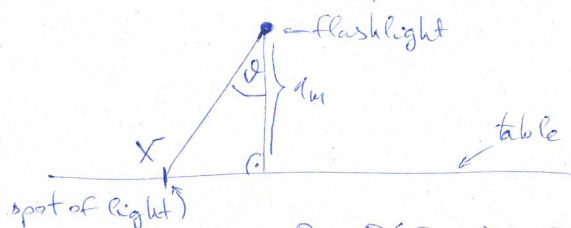
$X \in [-1, 1] \Rightarrow X^2 \in [0, 1]$

$$\textcircled{*} = P(|X| < \sqrt{y}) = P(-\sqrt{y} < X < \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \frac{1}{2}(\sqrt{y} - (-\sqrt{y})) = \sqrt{y}$$

$\frac{1}{2}$ $\leftarrow$ the density function of X

$$\Rightarrow \underline{f_Y(y)} = \begin{cases} 0, & y < 0 \\ \frac{1}{2} y^{-1/2}, & 0 \leq y < 1 \\ 0, & 1 \leq y \end{cases} \quad \text{is the density function of } X^2.$$

Example: Above the center of an infinite table a flashlight is placed, facing a random direction (but not upwards!). Assume it is placed only 1m above the table. We turn on the flashlight. What is the distribution of the position of the light spot on the table?



θ = angle of the flashlight
 X = position of the light spot
 $\theta \sim \text{Uni}[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $X = \tan \theta$

$$x \in \mathbb{R}: P(X \leq x) = P(\tan \theta \leq x) = P(\theta \leq \arctan(x)) = \frac{\arctan(x)}{\frac{\pi}{2}} + \frac{1}{2}$$

def. We say that a random variable X has Cauchy distribution with parameters θ and γ if it is absolutely continuous with ~~distribution~~ density function

$$f(x) = \frac{\gamma}{\pi((x-\theta)^2 + \gamma^2)} \quad \text{Notation: } X \sim \text{Cau}(\theta, \gamma)$$

In the previous example $X \sim \text{Cau}(0, 1)$.

Note: Variables with Cauchy distribution do not have expected value!

Prop. If $Y \sim \text{Cau}(0, 1)$ and $X = \gamma Y + \theta$, then $X \sim \text{Cau}(\theta, \gamma)$

proof:

$$P(Y \leq y) = \int_{-\infty}^y \frac{1}{\pi(1+x^2)} dx = \left[\frac{1}{\pi} \arctan(x) \right]_{-\infty}^y = \frac{1}{\pi} \arctan(y) + \frac{1}{2} = F_Y(y) \quad \leftarrow \text{the distribution functions}$$

$$P(X \leq x) = P(\gamma Y + \theta \leq x) = P(Y \leq \frac{x-\theta}{\gamma}) = \frac{1}{\pi} \arctan\left(\frac{x-\theta}{\gamma}\right) + \frac{1}{2} = F_X(x)$$

$$\Rightarrow f_X(x) = \frac{1}{\pi(1+(\frac{x-\theta}{\gamma})^2)} \cdot \frac{1}{\gamma} = \frac{\gamma}{\pi((x-\theta)^2 + \gamma^2)}$$

□